CHAPTER 3

Complex numbers and trigonometry

Addition formulas, exponential form, de Moivre's and Euler's formulas

Objectives. Establish the addition and double-angle formulas; write a complex number in exponential form and choose the form suited to a computation; compute powers; use de Moivre's and Euler's formulas to transform trigonometric expressions.

Prerequisites. Chapter 2; scalar product (Première); algebraic properties of the exponential function; unit circle and associated angles.

Fluency

Fluency · No calculator, five minutes.

1. Give $\cos\frac{\pi}{6}$ and $\sin\frac{5\pi}{6}$.
2. Express $\cos(\pi - x)$ and $\sin\left(\frac{\pi}{2} - x\right)$ in terms of $\cos x$ or $\sin x$.
3. Recall two expressions of the scalar product $\vec{a}\cdot\vec{b}$.
4. Simplify $\mathrm{e}^{a}\times\mathrm{e}^{b}$ and $(\mathrm{e}^{a})^{n}$.
5. Solve in $(-\pi,\pi]$ the equation $\cos x = \frac{1}{2}$.
6. Give the modulus and an argument of $-1-i$.

Solution · Fluency.

1. $\frac{\sqrt{3}}{2}$ and $\frac{1}{2}$.
2. $\cos(\pi - x) = -\cos x$ and $\sin\left(\frac{\pi}{2} - x\right) = \cos x$.
3. $\vec{a}\cdot\vec{b} = \left\|\vec{a}\right\|\left\|\vec{b}\right\|\cos\left(\vec{a},\vec{b}\right)$, and in an orthonormal frame $xx' + yy'$.
4. e^{a+b} and e^{na}.
5. $x = -\frac{\pi}{3}$ or $x = \frac{\pi}{3}$.
6. $\sqrt{2}$ and $-\frac{3\pi}{4}$.

Introductory activity

Activity 1 · A function that turns sums into products 35 min. For every real number θ, let $f(\theta) = \cos\theta + i\sin\theta$.

1. Compute $f(0)$, $f\left(\frac{\pi}{2}\right)$, $f(\pi)$. What is the modulus of $f(\theta)$? Where does the image of $f(\theta)$ lie?
2. Write $f\left(\frac{\pi}{3}\right)$ and $f\left(\frac{\pi}{6}\right)$ in algebraic form, then compute the product $f\left(\frac{\pi}{3}\right) f\left(\frac{\pi}{6}\right)$. Compare it with $f\left(\frac{\pi}{2}\right)$.
3. Compute likewise $f\left(\frac{\pi}{4}\right)^2$ and compare it with $f\left(\frac{\pi}{2}\right)$.
4. With a calculator or a spreadsheet, compare $f(a)f(b)$ and $f(a+b)$ for three pairs (a, b) chosen at random.
5. Which known function satisfies a relation of the same kind, $g(a)g(b) = g(a+b)$ and $g(0) = 1$?
6. Expand $f(a)f(b)$. Which trigonometric formulas would one need to know to prove the conjecture?

Solution · Activity.

1. $f(0) = 1$, $f\left(\frac{\pi}{2}\right) = i$, $f(\pi) = -1$. $|f(\theta)|^2 = \cos^2\theta + \sin^2\theta = 1$: the image lies on the circle with centre O and radius 1.
2. $f\left(\frac{\pi}{3}\right) = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ and $f\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2} + \frac{1}{2}i$. The product is $\frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} + i\left(\frac{1}{4} + \frac{3}{4}\right) = i = f\left(\frac{\pi}{2}\right)$.
3. $f\left(\frac{\pi}{4}\right)^2 = \left(\frac{\sqrt{2}}{2}(1+i)\right)^2 = \frac{1}{2} \times 2i = i = f\left(\frac{\pi}{2}\right)$.
4. The two numbers agree up to rounding errors. Conjecture: $f(a)f(b) = f(a+b)$.
5. The exponential function: $\mathrm{e}^a\mathrm{e}^b = \mathrm{e}^{a+b}$ and $\mathrm{e}^0 = 1$.
6. $f(a)f(b) = (\cos a\cos b - \sin a\sin b) + i(\sin a\cos b + \cos a\sin b)$. One needs to know that $\cos(a+b) = \cos a\cos b - \sin a\sin b$ and $\sin(a+b) = \sin a\cos b + \cos a\sin b$: these are the addition formulas, proved in the course.

Teacher note · Running the activity. Question 4 only brings numerical evidence; insist that it proves nothing. For quick students, one may add a remark beyond the syllabus: the "derivative" of f is $-\sin\theta + i\cos\theta = if(\theta)$, just as $\left(\mathrm{e}^{k\theta}\right)' = k\mathrm{e}^{k\theta}$ with $k = i$. This motivates the notation $\mathrm{e}^{i\theta}$.

Course notes

Addition and double-angle formulas

Property · Addition formulas. For all real numbers a and b:

$$\cos(a-b)=\cos a\cos b+\sin a\sin b, \qquad \cos(a+b)=\cos a\cos b-\sin a\sin b,$$

$$\sin(a-b)=\sin a\cos b-\cos a\sin b, \qquad \sin(a+b)=\sin a\cos b+\cos a\sin b.$$

Proof. In a direct orthonormal frame, let $\vec{p}\begin{pmatrix}\cos a\\ \sin a\end{pmatrix}$ and $\vec{q}\begin{pmatrix}\cos b\\ \sin b\end{pmatrix}$: these vectors are unit vectors and $(\vec{u},\vec{p})\equiv a, (\vec{u},\vec{q})\equiv b \pmod{2\pi}$. By Chasles' relation, $(\vec{q},\vec{p})\equiv a-b \pmod{2\pi}$. We compute $\vec{q}\cdot\vec{p}$ in two ways:

$$\vec{q}\cdot\vec{p}=\left\|\vec{q}\right\|\left\|\vec{p}\right\|\cos(a-b)=\cos(a-b) \quad \text{and} \quad \vec{q}\cdot\vec{p}=\cos b\cos a+\sin b\sin a.$$

This proves the first formula. Replacing b by $-b$ and using $\cos(-b)=\cos b$, $\sin(-b)=-\sin b$, we obtain $\cos(a+b)$. Finally $\sin(a+b)=\cos\left(\frac{\pi}{2}-a-b\right)=\cos\left(\left(\frac{\pi}{2}-a\right)-b\right)$, which the first formula expands into $\sin a\cos b+\cos a\sin b$; we deduce $\sin(a-b)$ by replacing b by $-b$. □

Property · Double-angle formulas. For every real number a: $\sin 2a=2\sin a\cos a$ and $\cos 2a=\cos^2 a-\sin^2 a=2\cos^2 a-1=1-2\sin^2 a$.

Example 1 · An exact value. $\cos\frac{\pi}{12}=\cos\left(\frac{\pi}{3}-\frac{\pi}{4}\right)=\frac{1}{2}\times\frac{\sqrt{2}}{2}+\frac{\sqrt{3}}{2}\times\frac{\sqrt{2}}{2}=\frac{\sqrt{6}+\sqrt{2}}{4}$.

Product and arguments

Property · Product of two trigonometric forms. For all real numbers θ and θ': $(\cos\theta+i\sin\theta)(\cos\theta'+i\sin\theta')=\cos(\theta+\theta')+i\sin(\theta+\theta')$.

Consequently, for all nonzero complex numbers z, z' and every integer n:

$$\arg(zz')\equiv\arg z+\arg z', \qquad \arg\frac{1}{z}\equiv-\arg z,$$

$$\arg\frac{z}{z'}\equiv\arg z-\arg z', \qquad \arg(z^n)\equiv n\arg z \pmod{2\pi}.$$

Proof. Expanding gives $(\cos\theta\cos\theta'-\sin\theta\sin\theta')+i(\sin\theta\cos\theta'+\cos\theta\sin\theta')$, which the addition formulas simplify. If $z=r(\cos\theta+i\sin\theta)$ and $z'=r'(\cos\theta'+i\sin\theta')$, then $zz'=rr'(\cos(\theta+\theta')+i\sin(\theta+\theta'))$ with $rr'>0$: this is a trigonometric form of zz'. The other relations follow: $z\times\frac{1}{z}=1$ has argument 0, then $\frac{z}{z'}=z\times\frac{1}{z'}$, and an induction for $n\geq 0$, extended to $n<0$ through the inverse. □

Key point · Interpretation. Multiplying by $\cos\theta+i\sin\theta$ turns through an angle θ about O. The conjecture of the activity of chapter 2 is proved: multiplying by i is a direct quarter turn, multiplying by $1+i=\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)$ is an eighth of a turn followed by an enlargement with ratio $\sqrt{2}$.

Exponential form

Definition · Imaginary exponential. For every real number θ, we write $e^{i\theta} = \cos\theta + i\sin\theta$. Every nonzero complex number can be written $z = re^{i\theta}$ with $r = |z|$ and $\theta \equiv \arg z \pmod{2\pi}$: this is its *exponential form*.

Property · Rules of computation. For all real numbers θ, θ' and every integer n:

$$e^{i\theta}e^{i\theta'} = e^{i(\theta+\theta')}, \qquad \frac{1}{e^{i\theta}} = e^{-i\theta} = \overline{e^{i\theta}}, \qquad \frac{e^{i\theta}}{e^{i\theta'}} = e^{i(\theta-\theta')}, \qquad \left(e^{i\theta}\right)^n = e^{in\theta}.$$

Values to know: $e^{i0} = 1$, $e^{i\pi/2} = i$, $e^{i\pi} = -1$, $e^{-i\pi/2} = -i$, $\left|e^{i\theta}\right| = 1$.

Property · De Moivre's formula. For every real number θ and every integer n: $(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta)$.

Property · Euler's formulas. For every real number θ: $\cos\theta = \dfrac{e^{i\theta} + e^{-i\theta}}{2}$ and $\sin\theta = \dfrac{e^{i\theta} - e^{-i\theta}}{2i}$.

Proof. $e^{i\theta} + e^{-i\theta} = e^{i\theta} + \overline{e^{i\theta}} = 2\,\mathrm{Re}\left(e^{i\theta}\right) = 2\cos\theta$, and $e^{i\theta} - e^{-i\theta} = 2i\,\mathrm{Im}\left(e^{i\theta}\right) = 2i\sin\theta$. □

Method · Choosing the suitable form. For a sum, the algebraic form; for a product, a quotient or a power, the exponential form. Example: $1 + i\sqrt{3} = 2e^{i\pi/3}$, so

$$\left(1 + i\sqrt{3}\right)^{10} = 2^{10}e^{10i\pi/3} = 1{,}024\,e^{-2i\pi/3} = -512 - 512\sqrt{3}\,i,$$

since $\dfrac{10\pi}{3} - 4\pi = -\dfrac{2\pi}{3}$.

Method · Expressing $\cos 3\theta$ in terms of $\cos\theta$: de Moivre. Expand $(\cos\theta + i\sin\theta)^3$ with the binomial formula and identify the real part: $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta$.

Method · Linearising $\cos^3\theta$: Euler. $\cos^3\theta = \dfrac{\left(e^{i\theta} + e^{-i\theta}\right)^3}{8} = \dfrac{e^{3i\theta} + 3e^{i\theta} + 3e^{-i\theta} + e^{-3i\theta}}{8} = \dfrac{\cos 3\theta + 3\cos\theta}{4}$.

Method · Half angle. To factorise a sum of exponentials, take out the exponential of the mean angle: $e^{ia} + e^{ib} = e^{i\frac{a+b}{2}}\left(e^{i\frac{a-b}{2}} + e^{-i\frac{a-b}{2}}\right) = 2\cos\dfrac{a-b}{2}\,e^{i\frac{a+b}{2}}$.

Watch out. $e^{i\theta}$ is not a positive real number: its modulus is 1, but it can be negative ($e^{i\pi} = -1$) or non-real. An expression $re^{i\theta}$ is an exponential form only if $r > 0$.

Exercises

Exercise 3.1 · Exponential forms. Write in exponential form: $-2+2i$; $-1-i\sqrt{3}$; $5i$; $\dfrac{1+i}{1-i}$.

Solution. $-2+2i = 2\sqrt{2}\,\mathrm{e}^{3i\pi/4}$; $-1-i\sqrt{3} = 2\mathrm{e}^{-2i\pi/3}$; $5i = 5\mathrm{e}^{i\pi/2}$; $\frac{1+i}{1-i} = \frac{\sqrt{2}\mathrm{e}^{i\pi/4}}{\sqrt{2}\mathrm{e}^{-i\pi/4}} = \mathrm{e}^{i\pi/2}$.

Exercise 3.2 · Two ways of writing the same product. Let $z_1 = \sqrt{3}+i$ and $z_2 = 1-i$.

1. Write z_1, z_2 and then $z_1 z_2$ in exponential form.
2. Write $z_1 z_2$ in algebraic form.
3. Deduce the exact values of $\cos\dfrac{\pi}{12}$ and $\sin\dfrac{\pi}{12}$.

Solution.

1. $z_1 = 2\mathrm{e}^{i\pi/6}$, $z_2 = \sqrt{2}\mathrm{e}^{-i\pi/4}$, $z_1 z_2 = 2\sqrt{2}\mathrm{e}^{-i\pi/12}$.
2. $z_1 z_2 = \sqrt{3} - \sqrt{3}i + i + 1 = \left(\sqrt{3}+1\right) + \left(1-\sqrt{3}\right)i$.
3. Identifying: $2\sqrt{2}\cos\left(-\frac{\pi}{12}\right) = \sqrt{3}+1$ and $2\sqrt{2}\sin\left(-\frac{\pi}{12}\right) = 1-\sqrt{3}$. So $\cos\frac{\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} = \frac{\sqrt{6}+\sqrt{2}}{4}$ and $\sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} = \frac{\sqrt{6}-\sqrt{2}}{4}$.

Exercise 3.3 · Powers.

1. Compute $(1-i)^8$ and $\left(\sqrt{3}+i\right)^{12}$.
2. Determine the natural numbers $n \geq 1$ for which $\left(\sqrt{3}+i\right)^n$ is real; positive real; purely imaginary. Give the smallest in each case.

Solution.

1. $(1-i)^8 = \left(\sqrt{2}\right)^8 \mathrm{e}^{-2i\pi} = 16$ and $\left(\sqrt{3}+i\right)^{12} = 2^{12}\mathrm{e}^{2i\pi} = 4{,}096$.
2. $\left(\sqrt{3}+i\right)^n = 2^n \mathrm{e}^{in\pi/6}$. It is real if and only if $\frac{n\pi}{6} \equiv 0 \pmod{\pi}$, that is n a multiple of 6; the smallest is 6. It is a positive real number if and only if n is a multiple of 12; the smallest is 12. It is purely imaginary if and only if $\frac{n\pi}{6} \equiv \frac{\pi}{2} \pmod{\pi}$, that is $n = 3+6k$; the smallest is 3.

Exercise 3.4 · De Moivre's formula. Using de Moivre's formula and the binomial formula, express $\cos 4\theta$ in terms of $\cos\theta$ only, and $\sin 4\theta$ in terms of $\cos\theta$ and $\sin\theta$.

Solution. With $c = \cos\theta$ and $s = \sin\theta$: $(c+is)^4 = c^4 + 4ic^3 s - 6c^2 s^2 - 4ics^3 + s^4$. The real part gives $\cos 4\theta = c^4 - 6c^2 s^2 + s^4$; with $s^2 = 1-c^2$, $\cos 4\theta = 8c^4 - 8c^2 + 1$. The imaginary part gives $\sin 4\theta = 4c^3 s - 4cs^3$.

Exercise 3.5 · Linearising in order to integrate.

1. Using Euler's formulas, prove that $\sin^3 x = \dfrac{3\sin x - \sin 3x}{4}$.
2. Deduce the value of $\displaystyle\int_0^{\pi} \sin^3 x \, \mathrm{d}x$.

Solution.

1. $\sin^3 x = \dfrac{\left(\mathrm{e}^{ix} - \mathrm{e}^{-ix}\right)^3}{(2i)^3} = \dfrac{\mathrm{e}^{3ix} - 3\mathrm{e}^{ix} + 3\mathrm{e}^{-ix} - \mathrm{e}^{-3ix}}{-8i} = \dfrac{2i\sin 3x - 6i\sin x}{-8i} = \dfrac{3\sin x - \sin 3x}{4}$.
2. An antiderivative is $-\frac{3}{4}\cos x + \frac{1}{12}\cos 3x$. So the integral equals $\left(\frac{3}{4} - \frac{1}{12}\right) - \left(-\frac{3}{4} + \frac{1}{12}\right) = \frac{4}{3}$.

Exercise 3.6 · Modulus and argument of $1 + \mathrm{e}^{i\theta}$.

1. Prove that for every real number θ, $1 + \mathrm{e}^{i\theta} = 2\cos\frac{\theta}{2}\,\mathrm{e}^{i\theta/2}$.
2. Deduce the modulus and an argument of $1 + \mathrm{e}^{i\theta}$ when $\theta \in (-\pi, \pi)$.
3. Application: modulus and argument of $1 + \mathrm{e}^{2i\pi/3}$, then a check with the algebraic form.

Solution.

1. $1 + \mathrm{e}^{i\theta} = \mathrm{e}^{i\theta/2}\left(\mathrm{e}^{-i\theta/2} + \mathrm{e}^{i\theta/2}\right) = 2\cos\frac{\theta}{2}\,\mathrm{e}^{i\theta/2}$ by Euler's formula.
2. For $\theta \in (-\pi, \pi)$, $\frac{\theta}{2} \in (-\frac{\pi}{2}, \frac{\pi}{2})$, so $\cos\frac{\theta}{2} > 0$. The expression is an exponential form: modulus $2\cos\frac{\theta}{2}$, argument $\frac{\theta}{2}$.
3. Modulus $2\cos\frac{\pi}{3} = 1$ and argument $\frac{\pi}{3}$. Indeed $1 + \mathrm{e}^{2i\pi/3} = 1 - \frac{1}{2} + \frac{\sqrt{3}}{2}i = \frac{1}{2} + \frac{\sqrt{3}}{2}i = \mathrm{e}^{i\pi/3}$.

Exercise 3.7 · A sum of cosines. Let θ be a real number that is not a multiple of 2π, and n a natural number.

1. Prove that $\displaystyle\sum_{k=0}^{n} \mathrm{e}^{ik\theta} = \frac{\mathrm{e}^{i(n+1)\theta} - 1}{\mathrm{e}^{i\theta} - 1}$.
2. Using the half-angle method, prove that this sum equals $\mathrm{e}^{in\theta/2}\dfrac{\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$.
3. Deduce $\displaystyle\sum_{k=0}^{n} \cos(k\theta) = \frac{\cos\frac{n\theta}{2}\,\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$, then check the formula for $n = 2$ and $\theta = \frac{\pi}{2}$.

Solution.

1. It is a geometric sum with common ratio $q = \mathrm{e}^{i\theta} \neq 1$ (since $\theta \not\equiv 0 \pmod{2\pi}$): $\sum_{k=0}^{n} q^k = \frac{q^{n+1}-1}{q-1}$, and $q^k = \mathrm{e}^{ik\theta}$.

2. For every real number t, $e^{it} - 1 = e^{it/2}\left(e^{it/2} - e^{-it/2}\right) = 2i \sin\frac{t}{2} e^{it/2}$. The quotient is therefore $\dfrac{2i \sin\frac{(n+1)\theta}{2} e^{i(n+1)\theta/2}}{2i \sin\frac{\theta}{2} e^{i\theta/2}} = e^{in\theta/2} \dfrac{\sin\frac{(n+1)\theta}{2}}{\sin\frac{\theta}{2}}$; the denominator is not zero because $\frac{\theta}{2}$ is not a multiple of π.

3. Take the real part of both sides. For $n = 2$ and $\theta = \frac{\pi}{2}$: $\cos 0 + \cos\frac{\pi}{2} + \cos\pi = 0$, and the formula gives $\dfrac{\cos\frac{\pi}{2} \sin\frac{3\pi}{4}}{\sin\frac{\pi}{4}} = 0$.

Problem

Problem · Superposing two oscillations. In physics, a simple voltage or sound can be written $t \mapsto A\cos(\omega t + \varphi)$. We show here that a sum $a\cos x + b\sin x$ can be put in this form. We study $g(x) = \sqrt{3}\cos x + \sin x$.

1. Write $\sqrt{3} - i$ in exponential form.
2. Prove that $g(x) = \mathrm{Re}\left(\left(\sqrt{3} - i\right)\mathrm{e}^{ix}\right)$, then that $g(x) = 2\cos\left(x - \frac{\pi}{6}\right)$.
3. Deduce the maximum and the minimum of g on $\mathbb{R}$, and the real numbers in $[0, 2\pi)$ where they are reached.
4. Solve in $[0, 2\pi)$ the equation $g(x) = \sqrt{2}$.
5. Generalise: let a and b be two real numbers, not both zero. Prove that there exist $R > 0$ and a real number φ such that, for every real number x, $a\cos x + b\sin x = R\cos(x - \varphi)$. Express R in terms of a and b.

Solution · Problem.

1. $\left|\sqrt{3} - i\right| = 2$, $\cos\theta = \frac{\sqrt{3}}{2}$, $\sin\theta = -\frac{1}{2}$: $\sqrt{3} - i = 2\mathrm{e}^{-i\pi/6}$.
2. $\left(\sqrt{3} - i\right)(\cos x + i\sin x) = \sqrt{3}\cos x + \sin x + i\left(\sqrt{3}\sin x - \cos x\right)$, whose real part is $g(x)$. Moreover $\left(\sqrt{3} - i\right)\mathrm{e}^{ix} = 2\mathrm{e}^{i(x - \pi/6)}$, whose real part is $2\cos\left(x - \frac{\pi}{6}\right)$.
3. The maximum is 2, reached for $x - \frac{\pi}{6} \equiv 0 \pmod{2\pi}$, that is $x = \frac{\pi}{6}$; the minimum is -2, reached for $x = \frac{7\pi}{6}$.
4. $\cos\left(x - \frac{\pi}{6}\right) = \frac{\sqrt{2}}{2}$ is equivalent to $x - \frac{\pi}{6} \equiv \frac{\pi}{4}$ or $x - \frac{\pi}{6} \equiv -\frac{\pi}{4} \pmod{2\pi}$, that is $x \equiv \frac{5\pi}{12}$ or $x \equiv -\frac{\pi}{12} \pmod{2\pi}$. In $[0, 2\pi)$: $x = \frac{5\pi}{12}$ or $x = \frac{23\pi}{12}$.
5. The complex number $a - ib$ is nonzero; write it $R\mathrm{e}^{-i\varphi}$ with $R = \sqrt{a^2 + b^2} > 0$. Then $a\cos x + b\sin x = \mathrm{Re}\left((a - ib)\mathrm{e}^{ix}\right) = \mathrm{Re}\left(R\mathrm{e}^{i(x - \varphi)}\right) = R\cos(x - \varphi)$.